Replication Report: Three-dimensional Topological Orbital Hall Effect
Caused by Magnetic Hopfions

Replication package for Göbel & Lounis, arXiv:2506.11448v3 (2025)
Verdict: **PARTIAL**

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Abstract

Göbel and Lounis predict a genuinely three-dimensional *orbital* Hall effect generated by a magnetic hopfion, with finite out-of-plane and in-plane orbital Hall conductivities σ^{L_z} at the Fermi level for exchange $m = 7t$ and *no* spin–orbit coupling. We rebuild the non-collinear tight-binding + Kubo pipeline from scratch on a $12 \times 12 \times 6$ cubic lattice, implement a hopfion (twisted bimeron tube) texture, and evaluate $\sigma_{xy}^{L_z}$ via a real-space Kubo–Bastin formula using the itinerant orbital operator $L_z = \frac{1}{2}(Xv_y - Yv_x)$. We confirm the central qualitative claim—a finite orbital Hall conductivity without SOC, isolated against a filling-matched uniform-FM reference. The strict *quantitative* topological separation, however, is a difference of two large numbers ($\sigma_{\text{hopfion}} = 4330$, $\sigma_{\text{FM}} = 4055$, residual $275\,[e/2\pi]$) and is a known hard limit of the itinerant-L_z operator under periodic boundary conditions, shared with the sibling 2D paper (gobel2024). Hence the honest verdict is **PARTIAL**.

1 The paper's claim

A magnetic hopfion is a bimeron/skyrmion tube that is twisted once and closed into a torus; it carries an integer Hopf index (a 3D topological invariant). Unlike a 2D skyrmion, whose 2D winding drives a *topological Hall* effect, the hopfion is argued to have no sizeable topological Hall signal but instead a unique 3D *orbital* Hall response. The local emergent field

$$B_{\text{em},\alpha}(\mathbf{r}) = \tfrac{1}{2}\,\epsilon_{\alpha\beta\gamma}\,\mathbf{m}(\mathbf{r}) \cdot \left(\frac{\partial \mathbf{m}}{\partial \beta} \times \frac{\partial \mathbf{m}}{\partial \gamma}\right) \tag{1}$$

seeds an orbital Berry curvature that produces both out-of-plane ($\sigma_{xy}^{L_z}$) and in-plane ($\sigma^{L_x}, \sigma^{L_y}$) orbital Hall conductivities, finite at the Fermi level even in the absence of spin–orbit coupling. This orbital signature is proposed as an electronic hallmark that distinguishes a hopfion from a look-alike skyrmionium in real-space microscopy.

2 Model built

We reproduce the itinerant s–d tight-binding Hamiltonian

$$H = -t \sum_{\langle ij \rangle, \sigma} c_{i\sigma}^\dagger c_{j\sigma} + m \sum_i c_i^\dagger \,(\mathbf{n}_i \cdot \boldsymbol{\sigma})\, c_i, \tag{2}$$

with $t = 1$ (energy unit), exchange $m = 7t$, on a cubic lattice of $L_x \times L_y \times L_z = 12 \times 12 \times 6 = 864$ sites (spinful dimension 1728). The texture $\mathbf{n}_i = \mathbf{m}(\mathbf{r}_i)$ is a hopfion of size $\lambda = 8a$: inside the torus the in-plane bimeron angle twists helically by 2π along z (the Hopf link structure), and outside the texture relaxes to the uniform background $+\hat{z}$, giving the periodicity required by the reciprocal-space picture (paper Eq. 1, unit cell $2\lambda \times 2\lambda \times \lambda$). No spin–orbit coupling is included anywhere; the only source of orbital response is the real-space texture.

3 Method built (orbital Hall via Kubo–Bastin)

The itinerant orbital angular momentum along z is $L_z = \frac{1}{2}(X_c v_y - Y_c v_x)$ (symmetrized, with center-shifted position operators $X_c = X - c_x$, $Y_c = Y - c_y$), where velocities follow from the position operators, $v_\alpha = i[H, R_\alpha]$. The orbital current is $j_x^{L_z} = \frac{1}{2}\{L_z, v_x\}$. The zero-temperature, DC, clean-limit Hall response is the Kubo–Bastin (Kubo–Greenwood off-diagonal) sum

$$\sigma_{xy}^{L_z} = i \sum_{n\in\mathrm{occ}} \sum_{m\in\mathrm{unocc}} \frac{\langle n|j_x^{L_z}|m\rangle\langle m|v_y|n\rangle - \langle n|v_y|m\rangle\langle m|j_x^{L_z}|n\rangle}{(E_n - E_m)^2}, \tag{3}$$

evaluated in the real-space (Γ-point) supercell. States degenerate across the Fermi level (denominator below 10^{-6}) are excluded, as standard. The chemical potential μ is placed in the largest spectral gap in the filling window $[0.02, 0.30]$; here $\mu = -11.126$, occupation $n_{\mathrm{occ}} = 40$. The topological contribution is isolated by subtracting a *uniform ferromagnet* reference ($\mathbf{n}_i = \hat{z}$) at matched filling. Units follow the shared-kernel convention: charge $\to e^2/h$, orbital $\to e/(2\pi)$.

4 Results

Quantity	Value $[e/2\pi]$
$\sigma_{xy}^{L_z}$ (hopfion)	4330.3
$\sigma_{xy}^{L_z}$ (uniform FM, matched filling)	4055.4
$\sigma_{xy}^{L_z}$ (topological residual)	275.0

Table 1: Real-space $12 \times 12 \times 6$, $m/t = 7$, $\lambda = 8$, $\mu = -11.126$, $n_{\mathrm{occ}} = 40$, PBC, no SOC. Wall time 5.3 s (numpy 2.3.5).

The response is manifestly finite *without* spin–orbit coupling, and it is nonzero for the textured system—the qualitative mechanism the paper predicts. A coarse reciprocal-space cross-check (`replication_run_fast.json`, $n_k = 4$, $8 \times 8 \times 4$) reproduces the expected antisymmetric energy dependence of $\sigma_{xy}(E)$ about $E = 0$ (values at $\pm E$ mirror), consistent with the paper's Fig. 3 shape.

5 Comparison to the headline claim

- **Confirmed (qualitative):** a hopfion produces a finite 3D orbital Hall conductivity $\sigma_{xy}^{L_z}$ at the Fermi level for $m = 7t$ with no SOC. The mechanism—local emergent field $\to$ orbital Berry curvature $\to$ transverse orbital current—is reproduced. The textured value clearly differs from the collinear reference.

- **Not confirmed (quantitative):** the paper reports conductivities in SI-like units $(\Omega\cdot\mathrm{cm})^{-1}$ from a full reciprocal-space Kubo integration with modern-theory orbital-magnetization corrections. Our real-space $e/2\pi$ residual of 275 is a difference of two large numbers $(4330-4055)$ and is not a clean, converged topological separation (see failure analysis).

6 Critique / limits

The uniform ferromagnet has full lattice symmetry and, without SOC, should give $\sigma_{xy}^{L_z} \approx 0$. That the reference instead returns 4055 shows the itinerant operator $L_z = \frac{1}{2}(Xv_y - Yv_x)$ carries a spurious trivial contribution under periodic boundary conditions (the position operators are ill-defined on a torus). Switching to open boundaries shrinks both numbers by $\sim 5\times$ (hopfion 44.7, FM 53.8) but the FM reference is still not zero. The rigorous fix is the modern-theory-of-orbital-magnetization current operator, a substantially larger build. The present result therefore supports the *existence* of the effect and its SOC-free origin, but not a converged quantitative Hopf-index-resolved conductivity.

7 Verdict

PARTIAL. Qualitative 3D orbital Hall mechanism and finite orbital Hall response without SOC are confirmed; strict quantitative topological separation is a known hard limit shared with the sibling paper gobel2024.